

ECHS FA-10 — Time-Series Surge Detection Report

Billing Spike Analytics · FY 2025-26

FA-10 | Time-Series Surge Detection

CLASSIFICATION	PERIOD	RECORDS ANALYSED	QUERY SETS
RESTRICTED	01-Apr-2025 to 31-Mar-2026	1,292,605 IPD 5,990,171 OPD Claims	4 Signals × 2 Tracks / 3,345 Hospitals

ECHS — Airawat Research Foundation

22 May 2026

ECHS Time-Series Surge Detection — FA-10

FY 2025-26 · 1 April 2025 – 31 March 2026

Report date: 22 May 2026 **Methodology:** Dual-track (IPD + OPD) time-series analysis using four independent statistical signals to detect hospitals with artificially manufactured billing spikes.

Note to reviewer: This report is produced by an automated screening system. Every flagged hospital is a *lead for investigation*, not a confirmed finding. A qualified auditor should review each case before any action is taken.

At a glance (composite tier): 137 hospitals **HIGH_RISK** | 437 **SUSPICIOUS** | 2,013 **MONITOR** | 758 **CLEAN** out of 3,345 hospitals analysed.

About This Report

This report presents the findings of **ECHS FA-10 — Time-Series Surge Detection**, a systematic screening exercise applied to all ECHS claims for FY 2025-26.

The analysis detects hospitals that show **statistically anomalous spikes in their monthly billing patterns** — sudden surges in claim volume or amount that are inconsistent with genuine patient-flow dynamics. Billing surges are a well-recognised fraud indicator: they arise when hospitals batch-process manufactured or inflated claims, exhaust budget allocations at period-ends, or coordinate admissions to meet revenue targets.

Two tracks are analysed independently:

- **IPD (Inpatient)** — hospital admissions (CI_PATIENT_TYPE = 'I')
- **OPD (Outpatient)** — outpatient visits (CI_PATIENT_TYPE = 'O')

Four signals are applied to each track:

Signal	Track	What It Measures	Flag Threshold
S1 — Monthly Volume Spike	IPD + OPD	Max z-score of monthly claim count	$z > 2.0$
S2 — Monthly Amount Spike	IPD + OPD	Max z-score of monthly approved amount	$z > 2.0$
S3 — Quarter-End Surge	IPD + OPD	% of annual claims in last 15 days of Jun/Sep/Dec/Mar	$> 20\%$
S4 — Month-End Week Surge	IPD + OPD	% of annual claims in last 7 days of any month	$> 30\%$
S5 — Peak-to-Trough Ratio	IPD + OPD	Peak month count ÷ trough month count	$> 5.0x$

Each hospital receives an independent **IPD score** (0–5) and **OPD score** (0–5). The **composite tier** is the higher of the two individual track tiers.

Dual-Track Scoring Methodology

Per-Track Tier Assignment

Track Score	Track Tier
4–5	HIGH_RISK
3	SUSPICIOUS
1–2	MONITOR
0	CLEAN

Composite tier = higher of `ipd_tier` and `opd_tier` (severity order: HIGH_RISK > SUSPICIOUS > MONITOR > CLEAN).

Guards and Filters

Guard	Rule
Minimum annual volume	≥ 10 claims per track per hospital
Z-score / PTT stability	S1, S2, S5 applied only where hospital has ≥ 6 active months

The Network-Wide Picture

The FY 2025-26 dataset covers **1,292,605 IPD claims** across **2,688 hospitals** and **5,990,171 OPD claims** across **3,161 hospitals** — **3,345 unique hospitals** in total.

Monthly Claim Distribution

Month	IPD Claims	OPD Claims	Total
Apr-25	101,434	481,883	583,317
May-25	105,037	502,914	607,951
Jun-25	102,741	490,979	593,720
Jul-25	109,725	534,487	644,212
Aug-25	107,835	501,653	609,488
Sep-25	112,013	524,813	636,826
Oct-25	106,308	494,484	600,792
Nov-25	107,364	500,632	607,996
Dec-25	115,209	525,590	640,799
Jan-26	105,929	499,867	605,796
Feb-26	106,769	498,586	605,355
Mar-26	112,241	434,283	546,524

The network-wide IPD peak occurred in **Dec-25** (115,209 claims). The trough was **Apr-25** (101,434 claims), giving a network-level peak-to-trough ratio of **1.1x**. Quarter-end months (June, September, December, March) consistently show elevated volumes in many hospitals — a pattern this analysis quantifies at the hospital level.

Part I — IPD Surge Analysis

Scope: 2,688 hospitals with ≥ 10 annual IPD claims in FY 2025-26.

IPD Signal Fire Rates

Signal	Description	Hospitals Flagged	% of Track
s1_vol_spike	Max monthly claim count z-score > 2.0	612	22.8%
s2_amt_spike	Max monthly approved amount z-score > 2.0	597	22.2%
s3_qtrend	Quarter-end concentration (Jun/Sep/Dec/Mar last 15 days) > 20%	536	19.9%
s4_monthend	Month-end week concentration (last 7 days of any month) > 30%	203	7.6%
s5_peak_trough	Peak-to-trough monthly count ratio > 5.0x	780	29.0%

IPD Tier Distribution

Tier	Hospitals	Share
HIGH_RISK	66	2.5%
SUSPICIOUS	204	7.6%
MONITOR	1,352	50.3%
CLEAN	1,066	39.7%

Top HIGH_RISK Hospitals — IPD Track

These hospitals scored 4–5 out of 5 on the IPD track, indicating multiple simultaneous time-series anomalies:

Hospital	City	State	Annual IPD Claims	Peak Month	Peak Count	Qtrend %	Monthen d %	Max Z (Vol)	PTT Ratio	Score
KAIZEN SUPER SPECIAL ITY HOS PITAL	THANE WEST	15	20	2025-12	6	40.0%	35.0%	2.20	6.0x	5
CENTRE FOR SIGHT - JODHPU R	JODHPU R	22	112	2026-02	19	24.1%	31.2%	2.02	19.0x	5
THE R J HUNJHU NUWAL A SANK ARA EYE HO SPITAL	NAVI MUMBAI	15	38	2025-06	7	21.1%	31.6%	2.13	7.0x	5

Hospital	City	State	Annual IPD Claims	Peak Month	Peak Count	Qtrend %	Monthen d %	Max Z (Vol)	PTT Ratio	Score
KRISHNA SUPER SPECIALITY HOSPITAL (A UNIT OF CNG)	JODHPUR	22	98	2026-03	16	26.5%	31.6%	2.06	5.3x	5
KHANDE LWAL N URSING HOME	DAUSA	22	26	2025-05	7	26.9%	34.6%	2.09	7.0x	5
Vasco (RC Mumbai)	Vasco-D a-Gama	6	30	2025-12	15	30.0%	33.3%	2.19	15.0x	5
ASG HO SPITAL PVT. LTD.	AMRITSAR	21	30	2026-02	7	23.3%	43.3%	2.13	7.0x	5
SHARP SIGHT CENTRE	NEW DELHI	34	40	2025-12	8	22.5%	35.0%	1.80	8.0x	4
Divya Prabha Eye Hospital	THIRUVANANTHAPURAM	13	41	2025-11	9	17.1%	41.5%	2.33	9.0x	4
DELHI HEART & LUNG INSTITUTE - NEW DELHI	NEW DELHI	34	36	2025-05	8	19.4%	30.6%	2.30	8.0x	4

Part II — OPD Surge Analysis

Scope: 3,161 hospitals with ≥ 10 annual OPD claims in FY 2025-26.

OPD Signal Fire Rates

Signal	Description	Hospitals Flagged	% of Track
s1_vol_spike	Max monthly claim count z-score > 2.0	704	22.3%
s2_amt_spike	Max monthly approved amount z-score > 2.0	798	25.2%
s3_qtrend	Quarter-end concentration (Jun/Sep/Dec/Mar last 15 days) > 20%	479	15.2%
s4_monthend	Month-end week concentration (last 7 days of any month) > 30%	220	7.0%

Signal	Description	Hospitals Flagged	% of Track
s5_peak_trough	Peak-to-trough monthly count ratio > 5.0x	968	30.6%

OPD Tier Distribution

Tier	Hospitals	Share
HIGH_RISK	72	2.3%
SUSPICIOUS	268	8.5%
MONITOR	1,552	49.1%
CLEAN	1,269	40.1%

Top HIGH_RISK Hospitals — OPD Track

These hospitals scored 4–5 out of 5 on the OPD track:

Hospital	City	State	Annual OPD Claims	Peak Month	Peak Count	Qtrend %	Monthend %	Max Z (Vol)	PTT Ratio	Score
LASER SIGHT INDIA PVT LTD	AGRA	26	174	2026-03	33	25.3%	33.3%	2.26	5.5x	5
Painavu	Idukki	13	82	2025-12	23	31.7%	47.6%	2.44	11.5x	5
YADAV DENTAL CLINIC	REWARI	8	918	2026-03	263	24.4%	35.8%	2.64	43.8x	5
EYE Q SUPER SPECIALITY EYE HOSPITALS - VADODARA	VADODARA	7	34	2025-08	8	26.5%	44.1%	2.06	8.0x	5
VISION PLUS EYE HOSPITAL - AMRITSAR, AJNALA RD	AMRITSAR	21	147	2026-03	29	24.5%	32.0%	2.50	7.2x	5
ANUPAM EYE HOSPITAL & RESEARCH CENTRE PVT LTD - JAIPUR	JAIPUR	22	24	2025-09	6	29.2%	33.3%	2.83	6.0x	5
T C EYE CARE	LUCKNOW	26	35	2026-03	8	25.7%	40.0%	2.36	8.0x	5

Hospital	City	State	Annual OPD Claims	Peak Month	Peak Count	Qtrend %	Monthend %	Max Z (Vol)	PTT Ratio	Score
LATHA EYE HOSPITAL	NAMAKKAL	24	36	2025-06	8	36.1%	47.2%	2.35	8.0x	5
HARIPRIYA DENTAL CLINIC	GUDUR, TIRUPATI	1	43	2025-09	16	23.3%	30.2%	2.25	16.0x	5
Iritty	Iritty	13	39	2025-06	7	30.8%	33.3%	2.07	7.0x	5

Part III — Composite Risk Tier

Composite Tier Distribution

Composite Tier	Hospitals	Share of Network
HIGH_RISK	137	4.1%
SUSPICIOUS	437	13.1%
MONITOR	2,013	60.2%
CLEAN	758	22.7%

Hospitals HIGH_RISK on Both IPD and OPD Tracks

These 1 hospitals are the highest-priority cases: they show simultaneous anomalous billing surges in both inpatient and outpatient streams, strongly suggesting a coordinated, institution-wide billing manipulation strategy rather than isolated noise.

Hospital	City	State	IPD Score	OPD Score	Total Score
EYE Q VISION PVT LTD - ALIGANJ	LUCKNOW	26	4	4	8

Key Observations and Recommendations

1. Composite HIGH_RISK Hospitals Require Immediate Audit

137 hospitals (4.1% of the network) are flagged as composite HIGH_RISK — meaning they triggered 4 or 5 signals on at least one track. These require field audit with ledger verification.

Recommendation: Prioritise field audit for all 137 HIGH_RISK hospitals. Request monthly admission registers and billing ledgers for the peak months identified in the signal tables above.

2. Dual-Track HIGH_RISK Hospitals Indicate Systemic Manipulation

1 hospitals show concurrent surge anomalies on both IPD and OPD tracks. This is inconsistent with genuine patient-flow variation and points to institution-level billing campaigns rather than coincidental spikes.

Recommendation: Place all dual-track HIGH_RISK hospitals on immediate payment hold pending audit. Cross-reference with FA-07 (emergency route misuse) data to identify overlap with established fraud patterns.

3. Quarter-End Surges Indicate Budget Manipulation

Signal S3 (quarter-end concentration) fired for **536** IPD hospitals and **479** OPD hospitals. These hospitals systematically concentrate claims in the last 15 days of June, September, December, and March.

Recommendation: Trigger automatic holds on claims filed in quarter-end windows for hospitals with a prior S3 flag. Require clinical justification for any month where a hospital's submission exceeds 20% of its annual total in a single 15-day window.

4. Month-End Week Surges Indicate Batch Claim Submission

Signal S4 (month-end week concentration) fired for **203** IPD hospitals and **220** OPD hospitals. These hospitals concentrate a disproportionate share of monthly admissions in the last 7 calendar days, consistent with batch-manufacturing of claims at period-end to meet billing quotas or exhaust budgets.

Recommendation: For hospitals with S4 flag, obtain admission registers for the final week of each flagged month and verify that each admission is supported by a corresponding clinical record. Particular attention should be given to hospitals where S4 and S3 co-fire (both month-end and quarter-end concentration).

5. Implement Real-Time Monthly Volume Alerts

Signals S1 and S2 detect months where a hospital's claim count or approved amount exceeds 2 standard deviations above its own baseline. This analysis is retrospective; the same logic should be applied monthly in real time.

Recommendation: Build a monthly automated alert that flags any hospital-month combination with a z-score above 2.0 on either volume or amount. Alert should trigger a desk review before the claims are approved for payment.

6. Cross-Reference Surge Months with Submission Batches

Peak months identified in the signal tables often correspond to large batch claim submissions rather than genuine admission surges. Comparing `CS_SUB_DATE` (submission date) against `CI_ADMISSION_DATE` will reveal whether hospitalisation and billing are temporally inconsistent.

Recommendation: For top HIGH_RISK hospitals, export all claims from their peak month and compute the gap between admission date and submission date. Gaps > 90 days, or submission date clustering on a single day, are strong evidence of batch-manufactured claims.

Limitations

- **Z-score with 12 monthly observations:** Standard deviation estimates are noisy with a small sample. The ≥ 6 -month activity guard reduces but does not eliminate false positives for hospitals with highly irregular claim patterns.
- **S5 (peak-to-trough) insensitivity at low volume:** For very low-volume hospitals, a single large claim in one month can produce an extreme PTT ratio even without fraud.
- **Unsubmitted claims excluded from S2:** `CS_UTI_APP_AMT` is NULL for claims not yet submitted to `claim_submission`; these are treated as 0 and may understate amount spikes for recently processed hospitals.
- **Hospital identity:** Join via `CI_CR_OFFICE_ID` → `OM_OFFICE_ID` — hospitals without a match in `office_master` are shown by ID only.
- **State shown as code:** `OM_OFFICE_STATE_ID` is a 3-character code; a join to `state_master` would yield full state names.
- **FY 2025-26 only:** A single year's data cannot distinguish seasonal patterns (e.g., eye surgery peaks before financial year-end) from manufactured surges. Multi-year trend analysis would strengthen findings.

Summary

Metric	Value
Total hospitals analysed (IPD track)	2,688

Metric	Value
Total hospitals analysed (OPD track)	3,161
Total hospitals in composite analysis	3,345
Total IPD claims examined	1,292,605
Total OPD claims examined	5,990,171
IPD HIGH_RISK hospitals (score 4–5)	66 (2.5%)
IPD SUSPICIOUS hospitals (score 3)	204 (7.6%)
OPD HIGH_RISK hospitals (score 4–5)	72 (2.3%)
OPD SUSPICIOUS hospitals (score 3)	268 (8.5%)
Composite HIGH_RISK (either track)	137 (4.1%)
Dual HIGH_RISK (both tracks)	1 (0.0%)
Network IPD peak month	Dec-25 (115,209 claims)
Network IPD peak-to-trough ratio	1.1x

Analysis performed by the CDIS Fraud Analytics team under project FA-10. Data source: ECHS transactional database, FY 2025-26. Raw data: `data/monthly\_fy2026.csv`. Report generated: 22 May 2026.